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Trust-Based Security Management in WSN-Assisted IoT Systems:
A Comprehensive Review

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ABSTRACT

The Internet of Things (IoT) with its pervasive connectivity imposes substantial challenges regarding privacy, security, and trust. Traditional security measures, such as cryptography and access control, often prove inadequate in the dynamic and resource-constrained IoT environment. The decentralized nature of IoT networks, coupled with the heterogeneity of devices, demands innovative security solutions. Trust is essential in human interaction and plays a crucial role in reducing challenges and security risks within IoT networks. This article focuses on trust-based methods for securing IoT systems, particularly within the context of Wireless Sensor Networks (WSNs). Trust-based approaches offer advantages such as lower computational complexity, leading to improved power management compared to traditional authentication and encryption methods. This article examines the latest trust models, as well as machine learning and deep learning algorithms specifically designed and implemented to enhance the security and reliability of trust-based WSN-assisted IoT applications.

KEYWORDS

Wireless Sensor Network (WSN), Internet of Things (IoT), Clustering, Routing, Metaheuristics algorithms, Swarm intelligence

INTRODUCTION

In 1999, Kevin Ashton, a British engineer and technology pioneer, coined the term “the Internet of Things (IoT)” to describe a system in which the Internet is linked to the physical world through ubiquitous sensors (Ashton, 2009). The International Telecommunication Union Telecommunication Standardization Sector (ITU-T) defines the IoT as “a global infrastructure for society, enabling improved services by interconnecting (physical and virtual) things based on existing and evolving interoperable information and communication technologies (Zmud et al., 2018).

IoT, a rapidly evolving technological phenomenon, is rapidly permeating our everyday lives. The sheer volume of interconnected devices, including everything from phones to appliances, is expected to reach approximately 40 billion globally by 2030 (Aaqib et al., 2023; Konsta et al., 2023), a figure

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poised for continuous year-over-year growth in the foreseeable future. Driven by a 15% annual growth rate, the IoT market shows an astonishing trajectory, skyrocketing from $1.90 billion in 2018 to an estimated $6 trillion by 2025 (Yalli et al., 2024). The dynamic market and widespread integration of IoT devices have unlocked numerous applications spanning diverse sectors (e.g., healthcare, agriculture, finance).

The IoT embeds intelligence into physical objects, driving new smart applications. However, these devices are constrained by computational power, energy consumption, and storage capacity, significantly hindering the implementation of effective on-device security (Kumar et al., 2023).

Wireless Sensor Networks (WSNs) are a key component of the IoT. These tiny sentinels are spread throughout our living environment, gathering real-time, valuable data used in every aspect of daily life (Fanian & Rafsanjani, 2025). The resulting WSN-based applications offer unique opportunities for improving lifestyles and introducing necessary new services that increase the overall quality of life. However, a major constraint of the sensors used in WSNs is their limited resources—such as battery life, storage capacity, and computation power (Saeedi et al., 2025). Therefore, energy consumption is one of the most critical parameters that must be considered.

IoT devices, which inherit constraints (e.g., limited battery and scalability) from WSNs and the internet, are increasingly susceptible to cyberattacks (Wang et al., 2021). Malicious actors exploit the widespread and varied deployment of these networks to access sensitive data without authorization (Okporokpo et al., 2023). Consequently, cyberattacks have significantly increased, including Social Engineering, Brute Force, Denial of Service (DoS), Man-in-the-Middle and Distributed Denial of Service (DDoS) attacks (Mekala et al., 2023). These attacks not only disrupt business processes and productivity but also pose severe threats to privacy and confidentiality.

The swift expansion of the Internet of Things, due to its dependence on internet connectivity, has significantly worsened issues related to security, privacy, and trust (Hammoudeh et al., 2020). Due to their diverse and extensive nature, IoT networks expand the threat landscape, risking the extraction of sensitive data. To counter this, robust security, privacy, and trust mechanisms must be implemented across all IoT layers, beginning with the physical devices (Adam et al., 2024).

The major challenges in these networks are limited energy and security. These challenges have led to efforts to improve energy management using more accurate clustering and routing techniques. Furthermore, several intelligent systems have been proposed to enhance security objectives from different aspects, often by utilizing clustering and routing techniques alongside them.

Despite extensive research in this field, our intention is to classify the trust-based articles mainly by their practical applications, specifically those related to the WSN-assisted IoT environment. Accordingly, we grouped the reviewed methods and schemes into four categories: clustering, routing, clustering/routing, and threat detection schemes. Furthermore, we performed further classification by considering the underlying algorithms, methods, or procedures applied for the implementation of these four categories. In addition, we have investigated the advantages and disadvantages of the methods/schemes

A comprehensive list of abbreviations employed throughout this paper, accompanied by their respective succinct definitions, is presented in the Appendix.

Contributions

This survey focuses on trust-based methods for enhancing the security and energy efficiency of clustering, routing, and combined clustering/routing protocols in WSN-assisted IoT applications. This literature review aims to classify and categorize the best trust-based research in these areas, presenting them in a concise and clear manner. Also, a sub-section is considered for trust-based threat-detection related articles in IoT applications. Furthermore, this survey will analyze the existing challenges and limitations of current trust-based approaches and discuss potential future research directions. The primary contribution of this review article is the classification of trust-based methods according to two concurrent aspects: their applications and the specific algorithms, methods, and procedures utilized for their implementation. Therefore, the contributions of this survey can be summarized as:

- Presentation of a new classification scheme that focuses on energy management and security in trust-based WSN-assisted IoT applications.
- Classification of various research articles and approaches into four distinct groups based on their main goals and methodologies (clustering, routing, clustering-routing and thread detection).
- Providing a general and comprehensive review of articles that have applied trust-based schemes to various IoT applications.
- Considering the co-classification of methods and procedures based on the auxiliary techniques and systems used in conjunction with primary trust management schemes (Fuzzy inference systems, Authentication and/or encryption, metaheuristics algorithms, and swarm intelligence algorithms).

The remainder of the paper is structured as follows: Section 2 reviews the existing survey and briefly compares them. Section 3 offers essential preliminaries; Section 4 presents a classification of trust-based methods according to various schemes; Section 5 discusses the reviewed articles, comparing and examining various aspects. Finally, Section 6 presents the conclusions and future directions.

EXISTING SURVEYS

There have been many studies on the concept of trust and its various applications in WSN-assisted IoT. Each of these surveys focuses on different aspects of trust, trust management, trust calculation, and more. Below, some of the recent and comprehensive related reviews and surveys are presented.

Hriez et al. (Hriez et al., 2021) focused primarily on well-known attacks within the IoT environment. They further investigated the concept of trust in the network and perception layers of IoT-enabled wireless sensor networks. For each layer, they reviewed several articles focusing on trust and attacks relevant to that specific layer.

Gautam and Kumar (Gautam & Kumar, 2021) examined the influence of key management, authentication, and trust management protocols on both the efficiency and security performance of wireless sensor networks. They conducted a detailed survey of the specifications and properties of various schemes addressing these issues, drawing from numerous articles. Within the section on trust management schemes, they focused primarily on distributed and centralized approaches.

Xia et al. (Xia et al., 2022) presented a review article, which examines the application of trust and reputation mechanisms to enhance the security of wireless sensor networks. In the first section, security concerns and established standards for WSNs are addressed, with a focus on confidentiality, data freshness, availability, authentication, integrity, forward and backward secrecy, access control, and non-repudiation. The second section addresses the vulnerability of WSNs to various threats and attacks, classifying them as internal/external and active/passive. Finally, after briefly defining and explaining the concept, types, and composition of trust, the article categorizes the reviewed studies into three application areas: trusted data fusion, trusted routing, and malicious node detection.

Tyagi et al. (Tyagi et al., 2023) discussed security and privacy challenges in IoT-based wireless sensor networks, along with existing solutions. Focusing on Trust Management Systems (TMS) as a key security approach, this review begins by examining the fundamental components of TMS, such as trust composition, indicators, evaluation, aggregation, propagation, and updating, highlighting their crucial role in safeguarding IoT security and privacy. Subsequently, the authors categorize trust management schemes into four groups: computational and probabilistic-based, cryptography-based, information theory-based, and other schemes. Finally, considering trust model components, common trust-based attacks in IoT, and desirable TMS characteristics, the survey refines its classification approach.

Shirvani and Masdari (Shirvani & Masdari, 2023) examined various techniques for trust management in Internet of Things (IoT) applications, focusing on a comprehensive study of trust issues in IoT security. Based on varying trust metrics, properties, and concepts, the authors categorized the reviewed articles into seven distinct groups: trust estimation, trust administration, trust-based access control, trust-based intrusion detection systems, trust-based data aggregation, trust-based RPL, and trust-based routing schemes. Finally, they analyzed the general aspects of the reviewed articles within this classification framework and explored common performance measures for trust-based IoT environments.

Konsta et al. (Konsta et al., 2023) conducted a systematic analysis and review of cutting-edge research in trust management approaches tailored for Internet of Things applications. The authors conducted an extensive and exhaustive review, categorizing the research into nine distinct categories based on various properties and specifications: blockchain-based, context-aware, social-based, game theory-based, probabilistic, prediction-based, fuzzy logic-based, direct trust, and recommendation-based approaches. These categories encompass different aspects of trust management, including information gathering, trust formation, and trust computation.

Alhandi et al. (Alhandi et al., 2023) presented an study which provides a comprehensive classification of trust models for Wireless Sensor Networks and Radio-Frequency Identification (RFID) networks, based on the methods employed. Initial sections explore the definition and characteristics of trust, along with existing trust model classifications from state-of-the-art surveys. The study then presents its own classification, considering trust attributes, trust architecture, trust aggregation model, distribution type of trust, model type, and the types of attacks the trust model can withstand. A more in-depth examination of trust evaluation models in WSNs and RFIDs follows. Finally, comparative tables offer a detailed evaluation of previous work, based on various trust evaluation characteristics.

Rajput and Yadav (Rajput & Yadav, 2025) examined recent articles on clustering protocols for IoT-enabled WSNs. It explored various schemes, including Machine Learning (ML), Deep Learning (DL), metaheuristics, hybrid optimization, trust-aware security, and energy harvesting, to classify the latest works in this field. By considering parameters such as network lifetime, energy consumption, latency, packet delivery ratio, throughput, and scalability, the paper presented a classification and evaluation of energy-efficient clustering protocols based on these parameters and the six aforementioned schemes. The review concluded that ML/DL and hybrid methods typically outperform metaheuristic approaches in terms of adaptability, lifetime, and reliability, despite their higher computational costs.

Wang et al. (Wang et al., 2025) presented a survey article that focuses on trust models and their effectiveness in detecting and defending against DoS attacks in WSNs. Unlike other relevant studies that investigate the capabilities of trust models for only two or three types of attacks, their review provides a comprehensive and comparative analysis of various types of DoS attacks. In their study, they examined essential trust evidence, approaches for defining trust evidence, and trust evaluation methods for developing efficient trust models. Additionally, they explored the impact of node authentication and link quality parameters on the trust evaluation model, as well as its application in network routing protocols.

A comparison table for examining the various aspects of the aforementioned survey and review articles is presented in Table 1.

Table 1. Summary of previous surveys on trust management.

Reference/Year	Trust Management Parameters	Security Measures	Trust Management Techniques/Categorization	Attacks	Contributions
(Hriez et al., 2021)	- Direct trust - Indirect trust - Watchdog mechanism	N/A	N/A	N/A	- Evaluate data quality in the perception layer, and assess watchdog mechanisms in the network layer.
(Gautam & Kumar, 2021)	- Distributed - Centralized	-Key management scheme - Authentication scheme - Trust Management scheme	- Distributed trust management schemes - Centralized trust management schemes	-HELLO Flood Attack - DoS Attacks - Sink Hole Attack - Wormhole Attack - Sybil Attack - Selective Forwarding	- Investigating systematic review for security solutions in WSNs. - Dividing key management, authentication and trust management schemes to different sub sections.
(Xia et al., 2022)	- Decentralized - Centralized	-Confidentiality - Availability - Forward and Backward Secrecy - Access Control and Nonrepudiation - Authentication	- Trusted data fusion - Trusted routing - Malicious node detection	- Internal and External Attacks - Passive and Active Attacks	- There is no contribution section by the Authors.
(Tyagi et al., 2023)	- Trust aggregation (TA) - Trust Composition (TC) - Trust computation - Trust Updation	- via Trust Management System (TMS)	- Cryptography based - Computational and Probabilistic based - Information Theory based - Other Methods	- On off attack - Sybil attack - DDOS - Wormhole attack, - Bad mouthing attack - Self promoting attack - Selfish attack	- A structured literature review on the importance of TMS for security and privacy of IoT. - Analyzed trust evaluation and management techniques. - Highlighted key characteristics of effective TMS.

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Table 1. Continued

Reference/Year	Trust Management Parameters	Security Measures	Trust Management Techniques/Categorization	Attacks	Contributions
(Shirvani & Masdari, 2023)	- Trust Composition - Trust Propagation (TP) - Trust Aggregation - Trust Update	N/A	- Trust Administration Schemes - Trust Estimation Schemes - Trust-Based Access Control - Trust-Based Data Aggregation - Trust-Based IDS Schemes - Trust-Based RPL Schemes - Trust-Based Routing Schemes	- Self-promoting attack - Bad-mouthing attack - Ballot stuffing attack - Whitewashing attack - Opportunistic service attack - Discriminatory attack	- Classification of proposed trust administration methods. - Comparative analysis of test management methods (multi-aspect.
(Konsta et al., 2023)	- Direct trust - Indirect trust - Distributed - Centralized	N/A	- Blockchain - Probabilistic - Fuzzy, - Context - Direct - Recommendations - Social - Game Theory - Prediction	- Eclipse attack - Replay attack - Node Capture attack - Sybil attack - Denial of Service attack - Spoofing attack - Wormhole attack	- Reviewed existing trust management techniques. - Categorized current literature based on trust management methods/tools. - Highlighted current challenges and future research directions.
(Alhandi et al., 2023)	- Attributes - Architecture - Aggregation - Distribution - Type	N/A	- Fuzzy Logic - Belief Theory - Bayesian Systems - Regression Analysis - Weighted Sum - Machine Learning	- Bad-mouthing attack - Selective Forwarding attack - Denial of Service attack - On-Off attack - Sybil attack	- Conducted a thorough literature review of trust evaluation models. - Analyzed classifications of trust models in WSN/RFID assisted IoT. - Identified challenges and future direction.

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Table 1. Continued

Reference/Year	Trust Management Parameters	Security Measures	Trust Management Techniques/Categorization	Attacks	Contributions
(Rajput & Yadav, 2025)	- Trust - Latency - Energy Efficiency - Security	N/A	- Meta Heuristic - Machine Learning - Deep Learning - Hybrid Optimization	N/A	- Influence of ML/DL methods on energy efficiency. - Investigation of challenges in AI based systems. - Explored performance of ML/DL and hybrid approaches.
(Wang et al., 2025)	- Communication Trust Evidence - Energy Trust Evidence - Data Trust Evidence	- DoS Attacks Only	- Dempster–Shafer - Arithmetic Mean - Weighted Average - Outlier Detection - Forgetting Curve - Intrusion Detection System	- Blackhole Attack - Selective Forwarding - Sinkhole Attack - Flooding Attack - Sybil Attack - Bad Mouthing Attack	- Comprehensive classification of DoS attacks types. - Optimal methods for trust evidence identification. - Specifying major challenges for defense against DoS attacks.
Our article	- Trust models - Trust composition - Trust aggregation - Trust update - Trust propagation	- via Trust Management Systems	- Clustering schemes - Routing schemes - Clustering and routing schemes - Threat detection schemes	N/A	- Classification of trust-based methods into four main groups. - Categorization of trust-based articles according to the methods and algorithms they employ. - Investigation of machine learning and deep learning trust-based methods.

PRELIMINARIES

This section offers an overview of the basic concepts and notions required for readers to gain a general understanding of the topics and concepts to be discussed.

Trust Definition, Challenges, and Path Forward

In the burgeoning domain of Wireless Sensor Networks within the IoT, trust signifies the degree of confidence that devices and users have in the reliability and veracity of information and services exchanged within the network. As IoT systems proliferate exponentially, establishing and maintaining trust becomes paramount for ensuring secure, efficient, and autonomous interactions among a diverse ecosystem of interconnected devices. The dynamic nature of these environments, characterized by the continuous emergence of new devices and users, necessitates the development of sophisticated trust management mechanisms that can adapt to evolving threats and facilitate seamless collaboration.

Trust models in WSN-based IoT can be broadly classified into two categories: direct and indirect evaluation methods. Direct trust relies on firsthand experiences and interactions between nodes, while indirect trust leverages recommendations and feedback from other nodes within the network (Mohammadi et al., 2019). A variety of approaches, including those incorporating fuzzy logic and machine learning techniques, have been proposed to effectively quantify and manage trust levels within the network. These models aim to address critical challenges such as maintaining trust over time, safeguarding data security, and ensuring scalability in large-scale IoT deployments (Aaqib et al., 2023).

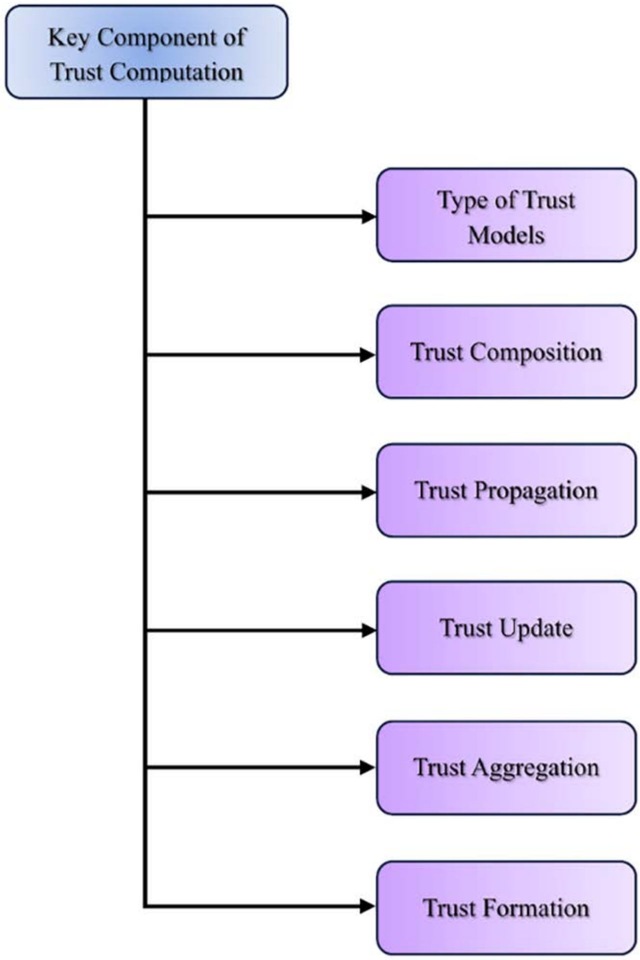
Despite significant advancements in trust management research, numerous challenges persist. The development of robust and efficient trust evaluation algorithms presents a significant technical hurdle. Moreover, the imperative to protect sensitive data from malicious entities necessitates the implementation of robust security measures. The implications of trust extend beyond technical considerations, profoundly influencing user perceptions and ultimately impacting the widespread adoption of IoT technologies (Alrahhal et al., 2022; Mohammadi et al., 2019).

Future research directions in trust management for WSN-based IoT will likely focus on enhancing existing trust frameworks, leveraging machine learning techniques for improved anomaly detection, and addressing the multifaceted cultural and regulatory factors that shape user trust perceptions. By prioritizing user needs and developing robust security protocols, the ultimate goal is to foster greater confidence in WSN-based IoT systems, thereby facilitating their seamless integration into everyday life (Ali-Eldin, 2022; Lata et al., 2021).

Key Components of Trust Computation Methods

Trust models are classified in (Kalidoss et al. 2020) according to six key aspects— trust models, composition, propagation, update, aggregation, and formation—with accompanying challenges identified. In Figure 1, the six key components of trust computation model are illustrated.

Figure 1. Key components of trust computation.



Types of Trust Models

For the Internet of Things, trust models are essential for establishing safe and dependable communication within WSNs (Gangwani et al., 2024). These models establish a framework for evaluating the trustworthiness of nodes, enabling them to collaborate effectively and share data securely (Hur et al., 2005). Broadly, there are three main categories of trust models employed in WSN-based IoT: centralized, distributed, and hybrid models.

Centralized trust models rely on a single authority to oversee trustworthiness, offering simplified management and improved interoperability, but they face challenges such as potential bottlenecks, single points of failure, and privacy concerns due to the central authority's extensive knowledge of network activities. In contrast, distributed trust models enable individual nodes to collaboratively assess trust, fostering autonomy and resilience; for instance, protocols like those developed by Chen et al. and Azad et al. utilize trust propagation and encryption techniques to enhance accuracy and privacy. Hybrid trust models aim to combine the strengths of both centralized and distributed approaches, leveraging the efficiency of centralized components while addressing their limitations through decentralized mechanisms, as demonstrated by (Tormo et al., 2015) adaptive model that optimizes trust evaluations based on real-time network dynamics.

Trust Composition

Trust Composition in WSN-assisted IoT applications involves determining which trust components should be considered and how they should be integrated to form a comprehensive trust assessment. Key components often include Quality of Service (QoS) trust and social trust (Guo et al., 2017). QoS trust evaluates factors such as data accuracy, delivery latency, and energy consumption, assessing the node's ability to fulfill its intended function (Chen et al., 2011). Social trust, on the other hand, reflects the node's trustworthiness within the social network, considering factors like reputation, past behavior, and recommendations from other nodes (Chen & Guo, 2014). By carefully selecting and combining these components, trust models can provide a more nuanced understanding of node trustworthiness, leading to more informed decision-making in routing, data aggregation, and other critical network operations. This approach enhances the resilience and reliability of WSN-assisted IoT systems by considering both functional performance and social behavior when evaluating and selecting trustworthy nodes for critical tasks.

Trust Propagation

In WSN-based IoT applications, trust propagation determines the dissemination of trust information among network nodes. Distributed propagation allows each node to independently assess and share trust information with its neighbors, promoting decentralization and resilience to single points of failure (Chen et al., 2014; Guo et al., 2017). However, it can lead to inconsistencies and information redundancy. Centralized propagation, on the other hand, relies on a central authority to collect, analyze, and disseminate trust information across the network (Saied et al., 2013). This approach can ensure consistency and accuracy but introduces a single point of vulnerability and may incur communication overhead. Several factors, including network size, communication limitations, and the needed level of decentralization and resilience, determine the proper scheme.

Trust Update

Is a mechanism in dynamic environments necessitate strategies for timely and accurate adjustments to trust values. Two prominent approaches are commonly employed: event-driven and time-driven schemes (Guo et al., 2017). Event-driven updates trigger modifications to trust values based on specific occurrences, such as successful data transmissions, observed anomalies, or reported malicious behavior. This approach offers reactivity to real-time events, enabling rapid adjustments to trust assessments (Chen et al., 2010). Conversely, time-driven updates periodically reassess trust values at predetermined intervals, regardless of whether significant events have occurred. While less reactive than event-driven updates, time-driven schemes ensure regular reevaluation of trust, preventing stale or inaccurate assessments in slowly evolving environments. The specific needs of the application determine which scheme should be chosen, such as the frequency and impact of expected changes in the environment, the computational resources available, and the desired level of responsiveness to trust-relevant events.

Trust Aggregation

In WSN-assisted IoT applications involves synthesizing diverse trust evidence gathered through self-observations or peer feedback to establish a comprehensive evaluation of a node's trustworthiness. This process is pivotal for informed decision-making in critical network operations, including data routing, resource allocation, and collaborative tasks (Guo et al., 2017). The optimal choice of aggregation technique depends on various factors, including the nature of trust information, the desired level of accuracy and robustness, computational resource constraints, and the specific requirements of the WSN-assisted IoT application.

Trust Formation

In WSN-based networks, trust formation involves establishing the initial level of trust between entities. This process often considers multiple trust properties, such as data accuracy, energy efficiency, and adherence to communication protocols. Research on trust formation commonly examines it through two distinct lenses: single-trust and multi-trust. Single-trust approaches focus on evaluating a single trust property, such as data accuracy, in isolation (Chen et al., 2014). Multi-trust approaches, on the other hand, consider multiple trust properties simultaneously, allowing for a more comprehensive and nuanced understanding of node trustworthiness (Chen & Guo, 2014). By integrating various trust properties, multi-trust approaches can provide a more robust and reliable foundation for secure and efficient operation in WSN-assisted IoT environments.

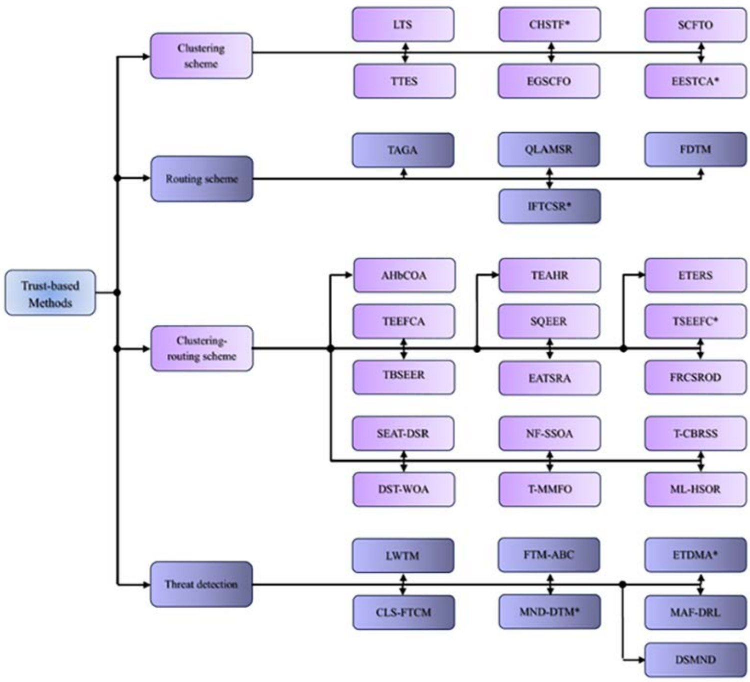
CLASSIFICATION OF TRUST-BASED METHODS

To analyze the breadth and depth of the research, this section categorizes the reviewed articles into four main groups based on their primary objectives. The groups include clustering schemes, which focus on methods for organizing network nodes into logical clusters; routing schemes, which investigate efficient data transmission strategies; integrated clustering and routing schemes, which combine these approaches to enhance network performance; and threat detection schemes, which aim to identify and mitigate security threats within the network environment. The classification of the selected articles is clearly depicted in Figure 2, which illustrates the categorization process based on the four above mentioned schemes. This figure provides a comprehensive overview of how trust-based articles are classified by its four major applications.

Clustering Schemes

Clustering in WSN-based IoT applications involves organizing sensor nodes into groups, where each group elects a cluster head (CH). This hierarchical structure enhances network scalability by reducing the number of direct transmissions to a base station (BS) (Manuel et al., 2020). Cluster heads aggregate data from their respective members, minimizing communication overhead and conserving energy. Clustering algorithms are designed to optimize energy usage throughout the network, thereby extending its operational lifespan. Main considerations include cluster head selection, inter-cluster communication, and adaptability to dynamic network conditions such as node failures and changes in environmental conditions. In this subsection, the trust-based articles with focus on clustering schemes are further investigated.

Figure 2. Classification of the trust-based methods based on the four primary applications.



- **Energy Efficient Secure Trust based Clustering Algorithm (EESTCA):** Rehman et al. (Rehman et al., 2017) proposed a method which uses different metrics for calculation of the weight of a node, which in used for secure CH selection. In the first step, for detecting misbehaving nodes, the direct trust evaluation of a node is done by considering parameters such as received packet rate ratio, successful packet sent rate, data forwarding rate and availability factor. Then, by use of the direct trust value of former step, the behavioral level of a node is specified and malicious node are identified. In the second step, by using the behavioral level, node waiting time, degree of node connectivity and relative mobility of a node, the cluster head selection algorithm is designed for secure and energy efficient clustering of sensor nodes in the network.
- **Large scale Trust Scheme (LTS):** Khan et al. (Khan et al., 2019) proposed a novel and comprehensive trust estimation approach (LTS) to improve the trustworthiness, security and cooperation between sensor nodes of WSN by detection of malicious nodes. The suggested approach operates in two level, the centralized inter-cluster and distributed intra-cluster scheme, for accurate trust estimation of sensor nodes with minimum overheads. Also, the trust severity and punishment can be tuned for better results, based on different application requirements.
- **TPM-based Trust Enhancement Scheme (TTES):** Wang et al. (Wang et al., 2019) proposed a trust enhancement scheme based on TPM hardware module for clustered WSNs. In setup phase of the suggested scheme, the proposed authentication protocols, SET-μTESLA and SET-SCHNORR, handle both cluster formation and the calculation of mutual trust relationships among cluster nodes. During the steady state, SET-μTESLA enhances energy efficiency by utilizing authenticated broadcast and minimizing hash function applications. Formal analysis confirms the proposed scheme's enhanced security against data confidentiality, data integrity, and compromised node attacks.

- **Secure Clustering protocol with Fuzzy Trust evaluation and Outlier detection (SCFTO):** Yang et al. (Yang et al., 2020) proposed a secure clustering protocol, designed to tackle transmission uncertainties in open wireless environments. This protocol utilizes Fuzzy Trust evaluation and Outlier detection. They employed a novel fuzzy logic controller of the interval type-2 variety for trust estimation. In the next step, by implementation of a density-based outlier detection method to establish an adaptive trust baseline, the identification and exclusion of malicious nodes to become a possible CHs are done. Furthermore, a fuzzy-based selection method for cluster heads, promoting energy efficiency while ensuring security. This approach increases the likelihood of normal sensor nodes with greater residual energy or more trust in compare to others, being selected as cluster heads.
- **Cluster Head Selection using Trust Function (CHSTF):** Narayan and Daniel (Narayan & Daniel, 2021) proposed a method which prolongs the network's service life and the remaining energy of the WSN's sensors In the first phase, the methods compute a new threshold value for CH selection, based on maximum residual energy and minimum distance of normal and advanced nodes to the base station. By utilizing the data fusion based on a notion of trust function, the preservation of data accuracy, enhancing QoS and minimization of data redundancy are done in the second phase.
- **Evolutionary Game based Secure Clustering protocol with Fuzzy trust evaluation and Outlier detection (EGSCFO):** Yang et al. (Yang et al., 2021) proposed an evolutionary game-based secure clustering protocol for WSNs, incorporating fuzzy trust evaluation and outlier detection. They begin by presenting a fuzzy trust evaluation method that converts transmission evidence into trust values, thereby reducing uncertainty in trust metrics. Subsequently, a K-Means based outlier detection plan analyzes the many trust values from fuzzy evaluations or recommendations. This method effectively identifies similarities and differences among sensor nodes, leading to more accurate outlier detection Finally, the protocol utilizes evolutionary game theory to balance security and energy efficiency during cluster head elections.

For further clarification of how trust-based methods are classified in this article, the two major groups of “Classification by Applications” and “Classification by Algorithms” are depicted in Figure 3.

Figure 3. The two major grouping of trust-based methods, “Classification by Applications” and “Classification by Algorithms”

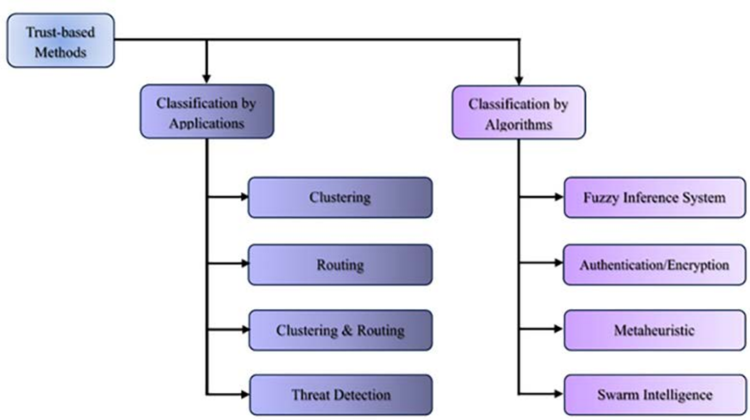


Table 2 presents a comparison and summary of clustering schemes and routing schemes, highlighting various aspects. In addition, advantages and disadvantages of clustering schemes are presented in Table 3.

Table 2. Comparison of discussed clustering schemes / methods.

Methods / Year	Authors	Used methods	Main features	Infrastructure network	Performance measures	Simulation environment
EESTCA (2017)	Rehman et al. (Rehman et al., 2017)	- Trust Evaluation - Behavioral Level Computation	Energy Efficient and Secure Trust Based Clustering in Mobile WSNs	MWSN	- Number of Compromised Nodes - Failure Rate - No. of Live Nodes - No. of Dead Nodes - Energy Consumption Ratio	Not mentioned
LTS (2019)	Khan et al. (Khan et al., 2019)	- Distributed Intra-Cluster Trust - Centralize Inter-Cluster Trust - Tune able Punishment and Trust Severity	- Trust-based Clustering Scheme in WSNs	WSN	- Severity Analysis - Communication Overhead - Malicious Node Detection - False Positive & Negative	MATLAB
TTES (2019)	Wang, et al. (Wang et al., 2019)	- μTESLA-based Broadcast - Authentication Protocol - Hardware TPM Module	Enhancement of Trust by using Authentication Protocols Improve Energy Efficiency	WSN	- Number of Alive Nodes - Network Life Cycle	Not mentioned
SCFTO (2020)	Yang et al. (Yang et al., 2020)	- Interval Type-2 Fuzzy Logic Controller - Density Based Clustering - Algorithm for Outlier Detection	Trust-based Secure Clustering and Outlier Detection in IWSN	IWSN	- No. of Dropping Attack - No. of Delaying Attack - Network Lifetime - Network Throughput	Not mentioned

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Table 2. Continued

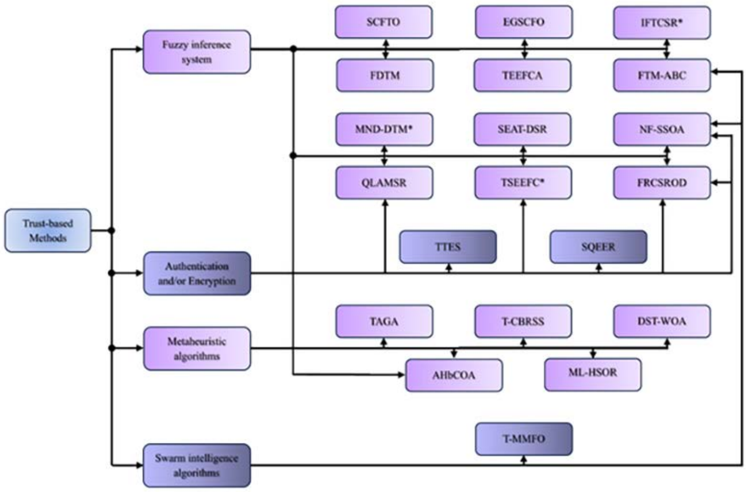
Methods / Year	Authors	Used methods	Main features	Infrastructure network	Performance measures	Simulation environment
CHSTF (2021)	Narayan and Daniel. (Narayan & Daniel, 2021)	- Trust Function - Two levels of energy for nodes	Trust-based Energy Efficient Clustering	WSN	- Number of Alive Nodes - Dead Nodes - Packet Sent - Energy Consumption - Death Rate - Remaining Energy	MATLAB
EGSCFO (2021)	Yang et al. (Yang et al., 2021)	-Interval Type-2 Fuzzy Trust Evaluation - K-Means based Outlier Detection - An Evolutionary Game Theory Approach to Secure Clustering	Secure Fuzzy-based Trust Evaluation and Clustering Outlier Detection	WSN	- Avg. No. of Malicious Cluster - No. of Dropping Attack - No. of Delaying Attack Network lifetime Network Throughput Timely Data Transfer??	MATLAB

Table 3. Advantages and disadvantages of clustering schemes / methods.

Methods / Reference	Pros	Cons
EESTCA (Rehman et al., 2017)	- Consideration of security in CH selection phase. - Never a malicious node could become a CH.	- Relatively computationally intensive.
LTS (Khan et al., 2019)	- Detection of selfish or faulty nodes. - Reliable path finding. - More security by introducing notion of TE.	- Storage and memory overhead. - More suitability for homogenous WSNs.
TTES (Wang et al., 2019)	- Using TPM module for detection of compromise attacks. - Introducing three effective and secure authentication protocols.	- Not applied and tested with real sensor environments. - Computational complexity.
SCFTO (Yang et al., 2020)	- Using Markov chain to model WSN channel quality. - Applying IT2-FLC for trust value estimation. - Proposes density based outlier detection scheme.	- No efficient transmission overhearing methods. - Complexity of data preparation mechanisms.
CHSTF (Narayan & Daniel, 2021)	- Applying weighted probability by SEP protocol for CH election. - Reducing unnecessary transmission energy in WSNs.	- Not all popular tests, such as PDR or throughput, were considered in the simulations.
EGSCFO (Yang et al., 2021)	- Using IT2 fuzzy logic for transform transmission parameters into trust values. - Applying K-Means for accurate malicious node detection.	- High computational complexity.

In Figure 4, another representation of different classification of the discussed papers, focusing base on the methods, algorithms, and techniques, is depicted.

Figure 4. Classification of trust-based methods with emphasis on the methods, algorithms, and techniques which are used



Routing Schemes

Routing in WSNs involves determining the optimal paths for data transmission between sensor nodes and the base station. This process is crucial for efficient data dissemination, ensuring timely and reliable information delivery while conserving energy (Ghaffari, 2014). Effective routing protocols consider factors such as node energy levels, network topology, traffic patterns, and application requirements. Key objectives include minimizing energy consumption, maximizing network lifetime, and ensuring data integrity and security (Narayana et al., 2024). Various routing protocols have been proposed, including data-centric routing, location-based routing, and hierarchical routing, each with its own strengths and weaknesses depending on the specific application and network characteristics.

- **Fuzzy-based Dynamic Trust Management for IoT (FDTM-IoT):** Hashemi and Shams Aliee. (Hashemi & Shams Aliee, 2020) proposed a method which utilizes the concept of trust as a comprehensive framework to implement countermeasures against the effects of attacks. Consequently, a multi-fuzzy, dynamic, and hierarchical trust model (FDTM-IoT) is introduced. This model encompasses three main dimensions: contextual information (CI), quality of service, and quality of peer-to-peer communication (QPC), each of which includes its own sub-dimensions or criteria. FDTM-IoT is integrated into the Routing Protocol for Low-Power and Lossy Networks (RPL) as an objective function, resulting in FDTM-RPL, which employs trust to address attacks. In this approach, fuzzy logic is applied in trust calculations to account for uncertainty, recognized as a crucial inherent characteristic of trust.
- **Trust-based Adaptive Genetic Algorithm (TAGA):** Han et al. (Han et al., 2022) proposed an energy-efficient, trust-based routing protocol, employing an adaptive genetic algorithm, for wireless sensor networks. This protocol aims not only to defend against common routing and specific trust attacks but also to reduce energy consumption associated with data transmission. TAGA determines the comprehensive trust values of nodes by processing their direct trust, which takes into account volatility and adaptive penalty factors, combined with indirect trust values obtained via filtering. A new threshold function is also introduced to identify the most suitable

cluster heads, by accounting for the dynamic fluctuations in nodes' comprehensive trust and remaining energy. Ultimately, an adaptive genetic algorithm is employed, featuring adjustable crossover and mutation probabilities, to determine the most secure routing for the cluster heads.

- **Query-based Location-Aware Secure Multicast Routing (QLAMSR):** Bhanu and Santhosh. (Bhanu & Santhosh, 2023) proposed a location-aware, query-based secure multicast routing protocol for wireless sensor networks. Its goal is to balance data integrity and energy consumption within network clusters while ensuring overall WSN security. The protocol is structured into three modules: The first defines the network model and system overview to enhance longevity. The second employs Advanced Encryption Standard (AES) and Rivest Cipher 6 (RC6) algorithms for authentication and data integrity. The third establishes energy-efficient routes, demonstrating network reliability. Additionally, a fuzzy decision model is implemented to assess node trust and the energy efficiency of reliable routes, thereby achieving effective data integrity.
- **Intuitionistic Fuzzy-based Trust Computation for Secure Routing (IFTCSR):** Mishra et al. (Mishra et al., 2023) proposed a trust-based model that encompasses various dimensions of trust, including direct trust, behavioral trust, and recommendations, aimed at addressing the issue of gap filling while considering dishonest recommendations from neighboring nodes and energy efficiency. This study introduced a secure route selection method for the Internet of Things environment by employing the widely recognized Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS). Furthermore, an intuitionistic fuzzy approach was applied to manage both numerical trust values and linguistic data. However, the dishonest suggestions from nearby nodes can render indirect recommendations vulnerable. The implementation of the intuitionistic fuzzy approach effectively addresses the challenges posed by dishonest advice.

Table 4 presents a comparison routing schemes, highlighting various aspects and Table 5 shows the advantage and disadvantage of routing schemes.

Table 4. Comparison of discussed routing schemes/methods

Methods/ Year	Authors	Used methods	Main features	Infrastructure network	Performance measures	Simulation environment
FDTM-IoT (2020)	Hashemi and Shams Aliee. (Hashemi & Shams Aliee, 2020)	- Fuzzy Inference System - RPL Protocol - Hierarchical Trust Evaluation	Efficient, Dynamic Fuzzy and Trust-based Routing Protocol for IoT	IoT	- Average Energy Consumption (AEC) - Average Number of Parent Changes - End-to-End Delay - Packet Loss Ratio	Cooja Simulator
TAGA (2022)	Han et al. (Han et al., 2022)	- Direct and Indirect Trust Value - Adaptive Genetic Algorithm	Energy Efficient and Secure Routing Protocol in WSN	WSN	Comprehensive Trust Value Lost Packets by Malicious Node Energy Consumption	MATLAB

continued on following page

Table 4. Continued

Methods/ Year	Authors	Used methods	Main features	Infrastructure network	Performance measures	Simulation environment
QLAMSR (2023)	Bhanu and Santhosh. (Bhanu & Santhosh, 2023)	- Fuzzy Inference System - Direct, Indirect and Route Trust - RC6 Encryption	Secure and Energy Aware Multicast Routing Protocol in WSNs	WSN	- End to end Delay - Network Lifetime - Detection Ratio - Encryption Time - Link Reliability Ratio	NS2
IFTCSR (2023)	Mishra et al. (Mishra et al., 2023)	- Trust Computation - Intuitionistic Fuzzy Inference System	Fuzzy-Based Trust Computation for Secure Routing in IoT	IoT	- Only Analytical Calculations	Mathematical analysis (No simulation)

Table 5. Advantages and disadvantages of routing schemes/methods

Methods / Reference	Pros	Cons
FDTM-IoT (Hashemi & Shams Aliee, 2020)	- Propose a trust model. - Applying the hierarchical fuzzy trust model.	- Not all common types of WSN attacks were considered and tested in the simulations. - High computational complexity.
TAGA (Han et al., 2022)	- Improving routing security (trust)and energy consumption. - Dynamically preventing the malicious nodes from becoming CHs.	- Common tests like PDR or network lifetime are not examined.
QLAMSR (Bhanu & Santhosh, 2023)	- Energy-efficient routing and re-route establishment. - Application of the Dijkstra algorithm for finding the shortest path based on hop count. - Balancing energy consumption and security via FIS.	- High complexity.
IFTCSR (Mishra et al., 2023)	- Proposing a novel fuzzy and trust-based method for evaluation of dependability in routing.	- Only the mathematical analysis of the suggested model is investigated, and no simulation is done.

CLUSTERING AND ROUTING SCHEMES

The subsequent section delves into scrutiny of research articles that employ trust-based methodologies within the context of WSNs. This analysis focuses on those studies that specifically integrate trust mechanisms for both clustering and routing protocols.

- **Energy Aware Trust Based Secure Routing Algorithm (EATSRA):** Selvi et al. (Selvi et al., 2019) proposed an energy-aware and trust-based secure routing algorithm for effective communication in wireless sensor network. The trust score is calculated in the initial phase, based on parameters such as packet delivery ratio, node behavior, and remaining energy. This score is then utilized to identify malicious nodes and find a secure paths for routing. In the second phase, by using the

Enhanced C4.5 decision tree algorithm the best path is selected from the secure paths discovered in the previous step.

- **Spatial and Energy Aware Trusted Dynamic Distance Source Routing (SEAT-DSR):** Mythili et al. (Mythili et al., 2019) proposed a method to prolong the lifespan of wireless sensor networks, by a spatial and energy-aware trusted dynamic distance source routing algorithm. This algorithm uniquely combines spatial information, energy levels, and data quality effectiveness using QoS-based energy-aware routing techniques. Additionally, a standard clustering algorithm is employed to group wireless sensor nodes based on trust scores, spatial data, energy levels, and inter-node distances. The SEAT-DSR algorithm is designed to make decisions based on specific evaluation metrics that reflect QoS. Furthermore, this model also introduces a new hierarchical trust mechanism that factors in several sensor node attributes, including data size, energy consumption, data communication speed and recommendations. Finally, a secure routing algorithm is presented, consisting of two stages: route discovery and route maintenance.
- **Secured QoS aware Energy Efficient Routing (SQEER):** Kalidoss et al. (Kalidoss et al., 2020) proposed a Secured Quality of Service aware Energy Efficient Routing Protocol based on the parameters of trust and energy, aiming for optimizing energy consumption and improve security in the network. To enhance the reliability of communication in this protocol, authentication and key based trust estimation modelling are used. Also, by considering direct and recommendation trust, the overall trust values (OTVs) and path trust scores (PTS) are computed and used for selection of best and most reliable routes, in routing section of the proposed protocol.
- **Fuzzy Rule and Cluster-based Secured Routing with Outlier Detection (FRCSROD):** Thangaramya et al. (Thangaramya et al., 2020), proposed a new cluster-based and fuzzy temporal rule-based routing algorithm that incorporates both trust modeling and outlier detection. Its primary aim is to monitor communication nodes and pinpoint malicious nodes within network clusters. The secure routing algorithm significantly improves routing decisions by employing temporal reasoning tasks, such as explanation-based learning and prediction, alongside spatial constraints. Through the application of trust and key management techniques, it performs robust node authentication, effectively isolating malicious nodes from communication via outlier detection.
- **Trust Based Secure and Energy Efficient Routing (TBSEER):** Hu et al. (Hu et al., 2021) proposed a method witch uses comprehensive trust value for an energy efficient routing and clustering in WSNs. With consideration of the indirect trust, direct trust and energy trust of nodes by base station for calculation of comprehensive trust value, further reduction in energy consumption is achieved in the network. Also, by introducing the notion of inspector nodes in the network for monitoring the behaviors of member and cluster head nodes, the selection of safe route through multiple paths could be done. Furthermore, by incorporating the adaptive penalty coefficient and volatilization factor, the malicious nodes are detected quickly.
- **Trusted Energy Efficient Fuzzy logic based Clustering Algorithm (TEEFCA):** Rajeswari et al. (Rajeswari et al., 2021) proposed a method for clustering and routing in WSN-based IoT, specifically designed to concurrently optimize security and energy conservation. This approach introduces a novel secure energy-aware cluster-based algorithm named Trusted Energy Efficient Fuzzy logic-based clustering Algorithm (TEEFCA) is introduced. By considering the notion of Node Fitness Value (NFV) the algorithm tries to compute trustworthy of candidate nodes and reduce the malicious node's chance. In the second phase, by considering the node residual energy, cluster density and distance between nodes and base station, the Fuzzy Inference System (FIS) is designed and applied for secure and stable clustering of nodes. The routing section of protocol is done by forming a routing table and selecting the nearest Cluster Leader (CL) node, among the other candidates, to transfer data from source to the sink.
- **Energy aware Trust-based Efficient Routing Scheme (ETERS):** Khan et al. (Khan et al., 2021) proposed a structured, trust estimation-based routing scheme for clustered wireless sensor

networks. It employs a multi-faceted trust approach—encompassing communication, energy, and data trust—to mitigate internal attacks such as badmouthing, Sybil, selective forwarding, on-off, black hole, and gray-hole attacks. To enhance the cluster head selection process, they introduce an innovative and efficient cluster head selection algorithm (ECHSA), which enables the election of a resilient CH at regular intervals, ensuring balanced load distribution among all CHs. ETERS leverages a Beta distribution-based trust function, as it allows for quicker recovery of trust values during attacks compared to Gaussian and Dirichlet distributions. Furthermore, a Trust-based Attack Detection Algorithm (TADA) is introduced. TADA assesses sensor node reliability to identify internal threats, using IDs, locations, and a triple trust metric. To address limitations in current trust models, the proposed system also includes an adjustable dynamic sliding length logical timing window. Furthermore, a trust-based secure routing algorithm is integrated to dynamically detect misbehavior in packet forwarding, fostering energy-efficient communication among sensor nodes.

- **Trust aware Secured Energy Efficient Fuzzy Clustering-based protocol (TSEEFC):** Kranthikumar and Velusamy. (Kranthikumar & Leela Velusamy, 2023) proposed a novel fuzzy-based and secured clustering and routing algorithm for improving security and energy efficiency of WSNs. By analyzing direct and indirect trust relationships between sensor nodes, alongside the energy consumption of neighboring nodes, it's possible to detect malicious nodes or energy depletion attacks. For more secure clustering of the sensor node, the base station provides the sensors with public and private keys. Then, the trust-based fuzzy method is used for cluster formation and based on the fuzzy rules generated, the routing phase is done for more secure and energy efficiency.
- **Trust-based Clustering and Best Route Selection Strategy (T-CBRSS):** Sudha and Tharini (Sudha & Tharini, 2023) proposed a method for trust-based and energy-efficient clustering and routing that utilizes Dempster-Shafer Theory and the Lion Optimization Algorithm (LOA). In the initial phase, the trust values of nodes are computed by evaluating two neighboring nodes, followed by the application of DST to determine the trustworthiness ratings of these nodes. In the subsequent phase, various trust metrics -including battery backup, packet forwarding capacity, distance, and latency- are considered to calculate the fitness value within the LOA framework, ultimately allowing for the selection of the most optimal route for data transmission.
- **Dempster-Shafer Theory with Whale Optimization Algorithm (DST-WOA):** Singh et al. (Singh et al., 2024) proposed a method that seeks to enhance the reliability and energy efficiency of wireless sensor networks through the seamless integration of trust management mechanisms into routing protocols. For trusted clustering, this method uses the Dempster-Shafer Theory's resilient decision-making algorithm, while the Whale Optimization Algorithm (WOA) handles routing. The main goal of this method is to improve the reliability and energy efficiency of WSNs by factoring in trustworthiness when selecting optimal clusters and routes.
- **Trust-based Modified Moth Flame Optimization (T-MMFO):** Shivakumaraswamy et al. (Shivakumaraswamy et al., 2024) proposed a trust-based modified moth flame optimization algorithm for secure cluster-based routing in wireless sensor networks. The T-MMFO algorithm is designed to select Secure Cluster Heads (SCHs) and routes to facilitate secure communication throughout the network, with the objectives of ensuring secure data transmission and maximizing the lifespan of the WSN. The fitness function for Cluster Head selection is calculated based on several parameters such as cluster density, inter-node distance, trust, remaining energy and traffic rate. Subsequently, the Moth Flame Optimization (MFO) algorithm is employed to identify the most secure and reliable nodes as CHs. In the following phase, T-MMFO utilizes a modified fitness function to select optimal secure routing paths.
- **Multi-Level Hierarchical Secure and Optimal Routing (ML-HSOR):** Sharma et al. (Sharma et al., 2024) introduced a multi-level hierarchical secure and optimal routing protocol designed to address various security threats posed by unauthorized intruders. This protocol features four

stages: registration, clustering, authentication, and optimal routing. During registration, new sensor nodes enroll with the base station using unique IDs. In the clustering stage, a Markov model with an adaptive weighting mechanism selects the most appropriate cluster head, extending the network's lifespan and improving performance. During the authentication phase, a multi-level trust evaluation process is used to pinpoint malicious nodes. Within this protocol, aggregated messages and their timestamps are encrypted, and the polarity learning-based chimp optimization algorithm (PL-COA) then establishes the most efficient data transmission path.

- **Neuro-Fuzzy-based clustering along with Sparrow Search Optimization Algorithm (NF-SSOA):** Dinesh and Santhosh Kumar (Dinesh & Santhosh Kumar, 2024) proposed a trust-aware neuro-fuzzy-based clustering method combined with a sparrow search optimization algorithm to facilitate energy-efficient, trust-aware, cluster-based secure data transmission in wireless sensor networks. The suggested system uses a neuro-fuzzy clustering algorithm for effective node clustering and a sparrow search optimization algorithm for routing. To secure communication, an ECC-based (Elliptic Curve Cryptography) digital signature algorithm is integrated, enabling efficient key generation, encryption, decryption, signature generation, and verification, along with hop-to-hop authentication for all WSN nodes. Furthermore, the proposed protocol incorporates pseudo-random identity generation to enable anonymous authentication during data transmission within the network.
- **Artificial Hummingbird Cheetah Optimizer Algorithm (AHbCOA):** Ramya and Brindha (Ramya & Brindha, 2025) The authors proposed a novel metaheuristic method called AHbCOA (Artificial Hummingbird Cheetah Optimizer Algorithm), which combines the Artificial Hummingbird Algorithm and the Cheetah Optimizer Algorithm to address the challenges associated with Cluster Head selection and routing in IoT-based Wireless Sensor Networks. In the first phase, multiple objectives such as energy, trust, and Low Latency Time (LLT) are considered, and predictions for these objectives are computed using a convolutional neural network trained with a cat-and-mouse-based optimizer model (CMBO). Next, the predicted objectives, along with Quality of Service (QoS) and distance parameters, are utilized in a fuzzy system for the CH selection phase. In the final phase, the AHbCOA algorithm is applied to determine the optimal routing path.
- **Trusted Energy-Aware Hierarchical Routing for Wireless Sensor Networks (TEAHR):** Vikas et. al. (Vikas et al., 2025) proposed a Trusted Energy-Aware Hierarchical Routing (TEAHR) method designed for multi-level trust evaluation and routing, which enhances the security level in WSNs. TEAHR employs various trust metrics, including energy trust, consistency trust, forwarding trust, behavioral trust, and anomaly detection. This comprehensive approach enables it to effectively tackle dynamic network topologies and emerging cyber threats. The method utilizes these parameters to compute a dynamic trust threshold, which is then used in the initial two steps of the cluster head selection process. In the final step, trust-based hierarchical routing is implemented through the application and assessment of different trust metrics and energy factors, resulting in secure data routing within the WSN.

Table 6 shows comparison and summary of clustering and routing schemes and Table 7 presents the advantages and disadvantages of the schemes.

Table 6. Clustering and routing schemes (methods) comparison

Methods/ Year	Authors	Used methods	Main features	Infrastructure network	Performance measures	Simulation environment
EATSRA (2019)	Selvi et al. (Selvi et al., 2019)	- Trust-based Malicious Node Detection - Enhanced C4.5 Decision Tree Algorithm	- Cluster-based and Energy-aware Secure Routing in WSN	WSN	- Delay Analysis - Packet Delivery - Energy Consumption - Malicious Node Detection Rate	NS2
SEAT-DSR (2019)	Mythili et al. (Mythili et al., 2019)	- Fuzzy Inference System - Dynamic Hierarchical Trust Mechanism	- Spatial and Energy Aware Clustering and Routing in WSN	WSN	- Transmission Rate - Throughput - Latency - Attack Detection Rate	NS2
SQEER (2020)	Kalidoss et al. (Kalidoss et al., 2020)	- Authentication - Direct Trust Score - Recommendation Trust Score	QoS Aware Secure Energy Efficient Clustering and Routing	WSN	- Throughput - Energy Consumption - Packet Delivery Ratio - Malicious Node Identification	NS2
FRCSROD (2020)	Thangaramya et al. (Thangaramya et al., 2020)	- Fuzzy Inference System - Advanced Encryption Standard	- Fuzzy Trust and Cluster-based Secure Routing in WSN - Outlier Detection	WSN	- Packet Delivery Rate - End-to-end delay - Throughput - Energy Consumption	NS2
TBSEER (2021)	Hu et al. (Hu et al., 2021)	- Adaptive Direct Trust - Indirect Trust - Energy Trust - Adaptive Penalty Mechanism	- Secure and Energy Efficient Clustering and Routing - Quick Malicious Node Detection	WSN	- Comprehensive Trust Value - Recognition Speed - Energy Consumption	MATLAB
TEEFCA (2021)	Rajeswari et al. (Rajeswari et al., 2021)	- Node Fitness Value - Trust Computation - Fuzzy Inference System	Secure and Energy-aware Trust-based Clustering and Routing	WSN-based IoT	- Number of alive nodes - Network lifetime - Average REL - Average Energy Consumed - False Positive Rate Analysis - Detection Accuracy - PDR, FND and LND	MATLAB
ETERS (2021)	Khan et al. (Khan et al., 2021)	- Energy, Data and Communication Trust - Beta Distribution - Trust based Attacks Detection Algorithm (TADA)	Secure Multi trust-based Routing algorithm in WSN	WSN	- Trust Value Severity - Malicious Node Detection - Network Throughput - Packet Delivery Rate (PDR) - Average Energy Consumption (AEC)	MATLAB
TSEEFC (2023)	Kranthikumar, and Velusamy. (Kranthikumar & Leela Velusamy, 2023)	- Authentication - Direct Trust - Indirect Trust - Fuzzy Inference System	Fuzzy-based Secure and Energy Efficient Clustering and Routing	WSN	- Packet Delivery Ratio - Network Lifetime - Communication Overhead - Computational Cost	NS2

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Table 6. Continued

Methods/ Year	Authors	Used methods	Main features	Infrastructure network	Performance measures	Simulation environment
T-CBRSS (2023)	Sudha and Tharini (Sudha & Tharini, 2023)	- Dempster-Shafer Theory - Lion Optimization Algorithm	- Trust-based Energy Efficient Clustering - Optimal and Energy Efficient Route Selection	WSN	- Packet Delivery Rate - Average Latency Rate - Energy Consumption - Network Lifetime	Not mentioned
DST-WOA (2024)	Singh et al. (Singh et al., 2024)	- Dempster-Shafer Theory - Whale Optimization Algorithm	- Trust-based and Energy Efficient Clustering and Routing in WSNs	WSN	- Energy Consumption - Throughput - Delay - Network Lifetime	MATLAB
T-MMFO (2024)	Shivakumaraswamy et al. (Shivakumaraswamy et al., 2024)	- Trust Computation - Moth Flame Optimization Algorithm	- Secure and Energy Efficient Clustering and Routing in WSNs	WSN	- Packet Delivery Ratio - Delay - Energy Consumption - Throughput - Network Lifetime	MATLAB
ML-HSOR (2024)	Sharma et al. (Sharma et al., 2024)	- Markov Model - Authentication by Direct and Indirect Trust - Polarity Learning-based Chimp Optimization Algorithm - Improved Blowfish Algorithm	- Multi-Level Trust-based Clustering and Secure Routing in IoT-WSN with resistance to some Attacks	WSN-based IoT	- Energy Consumption - Packet Delivery Ratio - End to End Delay - Network Lifetime - Throughput	Python 3.9
NF-SSOA (2024)	Dinesh and Santhosh Kumar (Dinesh & Santhosh Kumar, 2024)	- Enhance Energy Efficiency and Prolong Network Lifetime - Protection of Network Against Various Attacks	- Secure and Energy Efficient Clustering and Routing in WSNs	WSN	- Energy Consumption - Throughput - Network Lifetime - Network Delay - Packet Delivery Ratio	NS3
AHbCOA (2025)	Ramya and Brindha. (Ramya & Brindha, 2025)	- Fuzzy Inference System - Artificial Hummingbird - Cheetah Optimizer Algorithm - Convolutional Neural Networks - Cat-and-Mouse-Based Optimizer	- Fuzzy, Trust and DL-based Clustering and Routing in WSN	WSN-based IoT	- Energy Consumption - LLT - QoS - Trust Value - Throughput	Python
TEAHR (2025)	Vikas et. al. (Vikas et al., 2025)	- Multi-level trust evaluation - Hierarchical Routing	Trust and Energy-aware Clustering and Routing	WSN	- Average Latency - Packet Delivery Rate - Average Energy Consumption	Not mentioned

Table 7. Advantages and disadvantages of clustering & routing schemes/methods

Methods / Reference	Pros	Cons
EATSRA (Selvi et al., 2019)	- Applying the trust model for the detection of intruders in the network. - Employing the DST algorithm for secure route selection.	- Does not consider and handle incomplete data and parameters effectively. - Popular tests like throughput and lifetime are not examined.
SEAT-DSR (Mythili et al., 2019)	- Using various parameters and trust metrics to design the trust model. - Design a spatial and trust-based distance-aware secure routing protocol.	- No secure and fast data communication strategies.
SQEER (Kalidoss et al., 2020)	- Designing a secure and energy-efficient routing algorithm. - Applying spatial and temporal parameters for trust model design.	- Uncertainty of trust parameters are not considered in the dsigned protocol.
FRCSROD (Thangaramya et al., 2020)	- Design a security model based on FIS. - Proposing a fuzzy-based temporal, rule, and cluster-aware routing algorithm. - Discovery of malicious nodes using outlier detection.	- Intelligent agent-based schemes could be used for more effective data communications. - Complexity of FIS.
TBSEER (Hu et al., 2021)	- Introducing the notion of adaptive penalty coefficient for more accurate trust evaluation. - BS calculation of trust results in energy efficiency. - Using unspector nodes for monitoring and safe route selection.	- Popular tests like energy consumption and lifetime are not considered in simulations.
TEEFCA (Rajeswari et al., 2021)	- Introducing the NFV value for improving security and malicious node detection. - Enhancing network lifetime using FIS for routing.	- Complexity of the method due to multiple applications of fuzzy inference systems.
ETERS (Khan et al., 2021)	- Comprehensive multi-trust route construction and maintenance method. - Introducing a trust-based threat detection algorithm.	- Complexities in the routing algorithm subroutines.
TSEEFC (Kranthikumar & Leela Velusamy, 2023)	- Energy-efficient and secure fuzzy and cluster-based routing algorithm.	- Complexity of fuzzy inference systems and encryption/decryption algorithms.
T-CBRSS (Sudha & Tharini, 2023)	- Applying Dempster-Shafer theory for optimal CH selection. - Utilizing LOA for trust-based best route selection.	- Not enough details and information were mentioned about DST or LOA algorithms.
DST-WOA (Singh et al., 2024)	- Application of DST for trust-aware cluster formation. - Using WOA for optimized and robust data transmission and routing.	- Limited to homogenous networks and accuracy issues in dynamic environments. - Practical scalability and validation issues not considered.
T-MMFO (Shivakumaraswamy et al., 2024)	- Secure communications in WSNs using the modified MFO algorithm. - Using T-MMFOA for secure and attack-resistant route selection.	- High computational compelexity.

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Table 7. Continued

Methods / Reference	Pros	Cons
ML-HSOR (Sharma et al., 2024)	- Using a Markov model for real-time adjustment of parameters for optimal CH selection. - Significantly enhancing security and reliability using a multi-level trust scheme. - Congestion reduction using the PL-COA method.	- Scalability issues of the method for very large-scale networks. - Complexity of ML-HSOR routing algorithm.
NF-SSOA (Dinesh & Santhosh Kumar, 2024)	- Efficient cluster election by means of a neuro-fuzzy algorithm. - using ECC for assurance of authenticity and integrity of the model. - Optimal and energy-efficient routing using SSOA.	- Possibility of falling into a local optimum by the SSOA algorithm. - Data aggregation methods not applied for better communication quality.
AHbCOA (Ramya & Brindha, 2025)	- Application of the CNN model for the prediction of multiple parameters. - Using FIS for CH selection phase. - Combination of AHb and CO for the best path route finding.	- High computational compelexity. - The memory complexity of ML/DL methods is relatively high.
TEAHR (Vikas et al., 2025)	- Integration of trust metrics for optimal and energy-efficient routing.	- Computational complexity. - Energy consumption overhead.

Threat Detection Schemes

Threat detection is crucial for ensuring the security and reliability of WSNs. These networks are vulnerable to various threats, including node capture, data injection, eavesdropping, and denial-of-service attacks. Effective threat detection mechanisms are essential to identify and mitigate these threats in real-time. Common approaches include anomaly detection, intrusion detection systems, and reputation systems. Anomaly detection algorithms identify deviations from normal network behavior, while intrusion detection systems analyze network traffic for malicious activities. Reputation systems assess the trustworthiness of nodes based on their past behavior and interactions. By implementing robust threat detection mechanisms, WSN-based IoT applications can enhance their security, maintain data integrity, and ensure reliable operation in critical applications such as environmental monitoring and healthcare. Subsequently, this section delves deeper into research articles with primarily focus on threat detection techniques within the context of WSN-base IoT.

- **Lightweight Trust Mechanism (LWTM):** Singh et al (Singh et al., 2017) proposed a lightweight trust mechanism and overhead analysis for clustered WSN for protection of the system from different malicious attacks. By introducing the dynamic trust updating algorithm, they are used prioritized and divided metrics for sudden drop and gradual increase in node trust value. To encounter the misjudgment issue in the process of aggregation trust calculations, the self-adaptive weighted method is applied.
- **Fuzzy Trust Model and the ABC algorithm (FTM-ABC):** Pang et al. (Pang et al., 2021) proposed a method for detection of dishonest recommendation attacks in WSNs. In the first step, the Fuzzy Trust Model (FTM) is introduced to compute the direct and indirect trust between nodes of the WSN, by using the different communication, physical and data attributes. Then, the ABC algorithm is applied for detection and isolation of dishonest recommendation nodes. Furthermore, the security performance of the proposed method is improved by the utilization of recommended deviation and interaction index deviation fitness functions.

- **Efficient Trust-based Decision-Making Approach (ETDMA):** Khan et al. (Khan et al., 2023) proposed an efficient trust-based decision-making approach that employs a machine learning (ML) trust evaluation model tailored for unattended autonomous wireless sensor networks. This approach aims to enhance reliability, adaptability, scalability, and accuracy by swiftly generating dependable trust values in real time. To establish a comprehensive trust rating for sensor devices and anticipate potential misbehaviors, the proposed ML algorithm identifies several trust features: Co-Work Relationship (CWR), Cooperativeness-Frequency-Duration (CFD), Co-Location Relationship (CLR) and Reward (R). These trust features are integrated to produce a final trust rating, which informs decisions regarding the reliability of individual sensor devices.
- **Cross Layer Security Based on Fuzzy Trust Calculation Mechanism (CLS-FTCM):** Shang et al. (Shang et al., 2023) proposed a novel Cross-Layer Security approach based on a Fuzzy Trust Computing Mechanism, which optimizes deep learning for overhead monitoring in wireless sensor networks to effectively meet both storage and power requirements. Malicious nodes are identified using an optimized Bi-LSTM classifier within this framework. A novel self-improving bumblebee mating optimization (SI-BBMO) algorithm fine-tunes the Bi-LSTM model's weights to enhance attack detection accuracy. This is all enabled by a fuzzy logic-based cross-layer protocol that leverages inter-layer information to compute trust and mitigate WSN threats.
- **Malicious Node Detection using Dynamic Trust Management (MND-DTM):** Wang et al. (Wang et al., 2024) introduced a strategy for detecting malicious nodes based on dynamic trust management to tackle the challenges faced by Wireless Weak-link Sensor Networks (WWSNs). WWSNs are a specific category of wireless sensor networks which deployed in harsh and extreme environments, which results in fragile communication links between nodes, a phenomenon referred to as WWSNs by the authors. The dynamic trust management algorithm incorporates type-2 fuzzy logic and takes into account various trust factors to provide a comprehensive assessment of node trust within WWSNs. Furthermore, a mechanism for updating dynamic trust values is proposed to address the environmental changes that are inherent to WWSNs. Experimental results underscore the effectiveness of this approach in adapting dynamically to the network environment while maintaining a high level of performance in detecting malicious nodes.
- **Advancing Security and Trust in WSNs A Federated Multi-Agent Deep Reinforcement Learning Approach (MAF-DRL):** Moudoud et. al. (Moudoud et al., 2024) proposed a Multi-Agent Framework for Deep Reinforcement Learning (MAF-DRL) to address security threats that pose significant risks to the reliability and security of WSNs. Many AI-based IDSs schemes suffer from a lack of real-time and realistic attack data. To tackle this issue, they developed a method that combines Multi-Agent Deep Reinforcement Learning (MARL) with Federated Learning (FL). One of the main challenges of MARL is its slower training process and high consumption of network bandwidth. In contrast, privacy-aware collaborative methods like FL have emerged as a promising solution, enabling distributed model training across networks while preserving the privacy of each participant. Consequently, by implementing a trust-based mechanism that dynamically allocates resources based on the reliability of its agents, they designed a MAF-DRL system for adaptive, flexible, and robust attack detection.
- **Dual-stage machine learning approach for advanced malicious node detection in WSNs (DSMND):** Khashan (Khashan, 2025) proposed a novel Dual-Stage Malicious Node Detection (DSMND) scheme that utilizes machine learning techniques to enhance the identification rate of malicious nodes (MNs) in a WSN. In the first step, they implement an adaptive detection procedure using decision trees and dynamic threshold algorithms to optimize the selection of Cluster Heads based on varying resource levels, followed by the calculation of a threshold value for optimal MN identification. In the second step, when the threshold value is exceeded, an advanced hybrid server-side convolutional neural network combined with a random forest classifier model is employed for more accurate detection of MNs within the network.

Table 8 presents a comparison and summary of threat detection schemes, highlighting its various aspects and Table 9 indicates the advantages and disadvantages of the schemes.

Table 8. Comparison of threat detection schemes/methods

Methods/ Year	Authors	Used methods	Main features	Infrastructure network	Performance measures	Simulation environment
LWTM (2017)	Singh et al. (Singh et al., 2017)	- Inter and Intra-Cluster Trust - Self-adaptive Weight-based Trust Aggregation - Time Aging Factor	- Light-weight and Energy-efficient Malicious Node Detection	WSN	- Communication Overhead - Average Storage Overhead - Malicious Node Detection - False Positive and Negative	MATLAB
FTM-ABC (2021)	Pang et al. (Pang et al., 2021)	- Direct and Indirect Trust - Fuzzy Trust Model (FTM) - ABC Algorithm	-Fuzzy and Trust-based Malicious Node Detection	WSN	- Bad-Mouthing Attack - Ballot Stuffing Attack - Collision Attack - False Positive Rate	MATLAB
ETDMA* (2023)	Khan et al. (Khan et al., 2023)	- Real Time ML-based Trust Computation - Direct and Indirect Trust - K-Means Algorithm - PCA Analysis and SVM	- Real Time and Trust-based Decision-making approach for Threat and Attack Detection	WSN	- Accuracy - F1-score - Recall, FNR - Malicious Node Detection - Change in Trust Value	Not mentioned
CLS-FTCM (2023)	Shang et al. (Shang et al., 2023)	- Trust Computation - Fuzzy Inference System - Optimized Bi-LSTM classifier - Self-improving Bumblebee Mating Optimization Algorithm	- Security-based Fuzzy Trust Calculation for Threat Detection - Malicious Node Detection	WSN	- Malicious Node Detection Rate - False Positive Rate (FPR) - False Negative Rate (FNR) - Throughput - Energy Consumption	MATLAB
MND-DTM* (2024)	Wang et al. (Wang et al., 2024)	- Type-2 Fuzzy Inference System - Dynamic Trust Computation	- Malicious Node Detection in WWSNs	WWSN	- Recognition Accuracy based on Malicious Nodes Portion - Recognition Accuracy base on Rounds - Network Lifetime	MATLAB
MAF-DRL (2024)	Moudoud et. al. (Moudoud et al., 2024)	- Multi-Agent Deep Reinforcement Learning - Federated Learning - Graph Neural Networks	- Trust-based and Cluster-aware Threat Detection in WSNs	WSN	- Agent Rewards - Accuracy - True Positive Rate	Python
DSMND (2025)	Khashan (Khashan, 2025)	- Decision Tree Algorithm - Convolutional Neural Networks - Random Forest Classifier	- Malicious Node Detection in WSNs	WSN	- Accuracy - Execution Time - Energy Consumption - Network Lifetime - F1 Score - Recall	Cooja Simulator

Table 9. Advantages and disadvantages of threat detection schemes/methods

Methods / Reference	Pros	Cons
LWTM (Singh et al., 2017)	- Inter and intra-cluster based trust evaluation for more secure threat detection in WSNs.	- Various security attacks on the trust-based model are not investigated. - Relatively high memory overhead in some situations.
FTM-ABC (Pang et al., 2021)	- Comprehensive fuzzy-based indirect trust evaluation method. - Using the ABC algorithm for the detection of false recommendations and better malicious node detection.	- Adaptive threshold and parameters are not considered in the simulation. - High complexity.
ETDMA* (Khan et al., 2023)	- Use various trust metrics to train the model. - Applies the K-Means algorithm for the best cluster formation. - A combination of PCA nd SVM is used for dimensionality reduction.	- Convergence and scalability of the method. - High complexity of ML/DL methods.
CLS-FTCM (Shang et al., 2023)	- Application of Bi-LSTM for accurate malicious node identification. - Combining FIS and DL for more precise threat detection.	- Higher complexity of fuzzy inference systems and DL methods.
MND-DTM* (Wang et al., 2024)	- Investigate the security of the weak link in WSNs. - Dynamic mechanism of the designed method for malicious node detection using fuzzy type-2 sets.	- The performance of the proposed method has not been investigated in real-world scenarios. - High complexity.
MAF-DRL (Moudoud et al., 2024)	- Combining federated learning and reinforcement learning for threat detection in WSNs. - Novel behavior-based clustering scheme.	- Roubustness of method against adverserial attacks.
DSMND (Khashan, 2025)	- Applying a hybrid ML model for rapid malicious node detection at the CH level. - Combining CNN and RF for further refinement of the threat detection process at the server side.	- The adaptability of the model on different network topologies.

DISCUSSION

In this section, Figure 5 depicts the timeline diagram of the reviewed articles. Also, all the reviewed articles are compared in Table 10, which examines various aspects of WSNs, including methods, routing techniques, and commonly used performance evaluations for each method or scheme.

Figure 5. The timeline diagram of the reviewed schemes

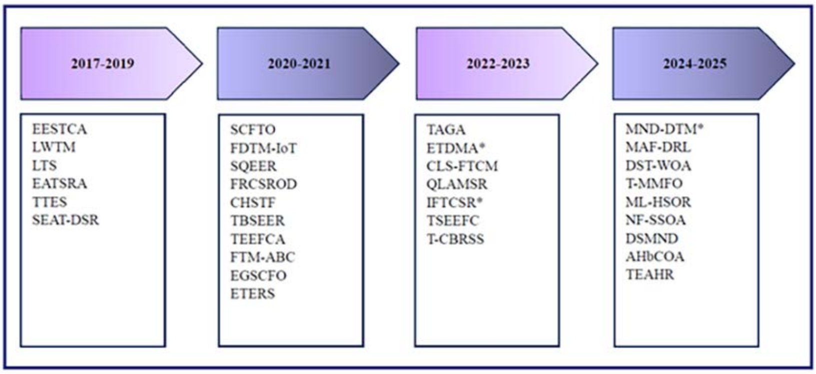

Table 10. Comparison of discussed methods/schemes based on various criteria and aspects

Method / Reference	Network Type		Application Type				Routing				Packet Delivery Ratio	Network Lifetime	Security	Accuracy	Energy Consumption	Latency / Delay
							Inter-		Intra-							
	Heterogenous	Homogenous	Clustering	Routing	Clustering & Routing	Threat Detection	Single Hop	Multi Hop	Single Hop	Multi Hop						
EESTCA (Rehman et al., 2017)	✗	✓	✓	✗	✗	✗	✗	✗	✗	✗	✗	✓	✗	✗	✓	✗
LWTM (Singh et al., 2017)	✗	✓	✗	✗	✗	✓	✗	✗	✗	✗	✗	✗	✓	✓	✗	✗
LTS (Khan et al., 2019)	✗	✓	✓	✗	✗	✗	✗	✗	✗	✗	✗	✗	✓	✓	✗	✗
EATSRA (Selvi et al., 2019)	✗	✓	✗	✗	✓	✗	✓	✗	✗	✓	✗	✗	✗	✓	✓	✓
TTES (Wang et al., 2019)	✗	✓	✓	✗	✗	✗	✗	✗	✗	✗	✗	✓	✓	✗	✓	✗
SEAT-DSR (Mythili et al., 2019)	✗	✓	✗	✗	✓	✗	✓	✗	✓	✓	✓	✓	✓	✗	✓	✓
SCFTO (Yang et al., 2020)	✗	✓	✓	✗	✗	✗	✗	✗	✗	✗	✗	✓	✓	✗	✓	✗
FDTM-IoT (Hashemi & Shams Aliee, 2020)	✗	✓	✗	✓	✗	✗	✗	✗	✗	✓	✗	✓	✗	✓	✓	✓
SQEER (Kalidoss et al., 2020)	✗	✓	✗	✗	✓	✗	✗	✗	✗	✓	✓	✗	✓	✓	✓	✗
FRCSROD (Thangaramya et al., 2020)	✗	✓	✗	✗	✓	✗	✗	✗	✗	✓	✓	✗	✓	✗	✓	✓
CHSTF (Narayan & Daniel, 2021)	✗	✓	✓	✗	✗	✗	✗	✗	✗	✗	✓	✓	✗	✗	✓	✗
TBSEER (Hu et al., 2021)	✗	✓	✗	✗	✓	✗	✓	✗	✗	✓	✗	✗	✗	✗	✓	✓
TEEFCA (Rajeswari et al., 2021)	✗	✓	✗	✗	✓	✗	✓	✗	✗	✓	✓	✓	✗	✓	✓	✗

continued on following page

Table 10. Continued

Method / Reference	Network Type		Application Type				Routing				Packet Delivery Ratio	Network Lifetime	Security	Accuracy	Energy Consumption	Latency / Delay
							Inter-		Intra-							
	Heterogenous	Homogenous	Clustering	Routing	Clustering & Routing	Threat Detection	Single Hop	Multi Hop	Single Hop	Multi Hop						
FTM-ABC (Pang et al., 2021)	✗	✓	✗	✗	✗	✓	✗	✗	✗	✗	✗	✗	✓	✓	✗	✗
EGSCFO (Yang et al., 2021)	✗	✓	✓	✗	✗	✗	✗	✗	✗	✗	✗	✓	✓	✓	✓	✗
ETERS (Khan et al., 2021)	✗	✓	✗	✗	✓	✗	✓	✗	✓	✓	✓	✗	✗	✓	✓	✓
TAGA (Han et al., 2022)	✗	✓	✗	✓	✗	✗	✓	✗	✓	✓	✗	✗	✗	✓	✓	✗
ETDMA* (Khan et al., 2023)	✗	✓	✗	✗	✗	✓	✗	✗	✗	✗	✗	✗	✗	✓	✗	✗
CLS-FTCM (Shang et al., 2023)	✗	✓	✗	✗	✗	✓	✗	✗	✗	✗	✗	✗	✓	✓	✓	✗
QLAMSR (Bhanu & Santhosh, 2023)	✗	✓	✗	✓	✗	✗	✓	✗	✗	✓	✓	✓	✗	✓	✓	✓
IFTCSR* (Mishra et al., 2023)	✗	✓	✗	✓	✗	✗	✗	✗	✗	✓	✓	✗	✓	✗	✗	✗
TSEEFC (Kranthikumar & Leela Velusamy, 2023)	✗	✓	✗	✗	✓	✗	✓	✗	✗	✗	✓	✓	✗	✗	✓	✓
T-CBRSS (Sudha & Tharini, 2023)	✗	✓	✗	✗	✓	✗	✗	✗	✗	✓	✓	✓	✗	✗	✓	✓
MND-DTM* (Wang et al., 2024)	✗	✓	✗	✗	✗	✓	✗	✗	✗	✗	✗	✓	✗	✓	✓	✗
MAF-DRL (Moudoud et al., 2024)	✗	✓	✗	✗	✗	✓	✗	✗	✗	✗	✗	✗	✓	✓	✗	✗
DST-WOA (Singh et al., 2024)	✗	✓	✗	✗	✓	✗	✗	✗	✗	✓	✗	✓	✗	✗	✓	✓
T-MMFO (Shivakumaraswamy et al., 2024)	✗	✓	✗	✗	✓	✗	✗	✗	✗	✓	✓	✓	✗	✗	✓	✓
ML-HSOR (Sharma et al., 2024)	✗	✓	✗	✗	✓	✗	✗	✗	✗	✗	✓	✓	✗	✗	✓	✓
NF-SSOA (Dinesh & Santhosh Kumar, 2024)	✗	✓	✗	✗	✓	✗	✗	✗	✗	✓	✓	✓	✓	✗	✓	✓
DSMND (Khashan, 2025)	✗	✓	✗	✗	✗	✓	✗	✗	✗	✗	✗	✓	✗	✓	✓	✗
AHbCOA (Ramya & Brindha, 2025)	✗	✓	✗	✗	✓	✗	✗	✗	✗	✓	✓	✗	✗	✗	✓	✗
TEAHR (Vikas et al., 2025)	✗	✓	✗	✗	✓	✗	✗	✗	✗	✓	✓	✗	✓	✗	✓	✓

CONCLUSION AND FUTURE DIRECTIONS

Energy efficiency and security represent significant research challenges in WSN-assisted IoT applications, particularly in the context of trust management and computation where nodes operate under limited energy constraints. Given the rapid advancement of IoT technology, there is an urgent need to enhance energy efficiency to accommodate the burgeoning variety of heterogeneous IoT devices. This study aims to explore various solutions for implementing trust management that preserves energy efficiency while balancing the security of IoT systems. The classification of these solutions is organized into four primary categories: clustering, routing, a combination of clustering and routing, and threat detection. Within each category, the most notable works are selected and elaborated upon in detail, providing readers with a comprehensive understanding of the research conducted in trust management and trust computation concerning energy efficiency and security in WSN-assisted IoT networks. Many existing trust models employ complex procedures and algorithms to calculate trust values. Therefore, there is a pressing need to develop a lightweight and straightforward trust model that minimizes resource consumption, including low memory usage, reduced processor demands, and lower power requirements. Furthermore, IoT devices are continually vulnerable to various types of attacks across different IoT layers. Consequently, the trust model must incorporate methods for trust computation that can effectively detect these diverse attack types within the IoT framework.

In recent years, there has been an increase in ML/DL/Federated learning (FL) based methods that attempt to improve the energy consumption, and security objectives of WSN-based IoT applications. This surge reflects the increasing recognition of artificial intelligence's potential to enhance network efficiency and robustness. However, these systems inherently face significant technical limitations, particularly regarding constrained energy and computation power within the physical sensors. This disparity between potential and constraint creates vast research opportunities for investigating the practical application of DL/ML/FL-based methods, specifically within trust-based WSN-assisted IoT environments. Future research efforts should be directed toward reducing energy consumption, increasing network lifetime, and rigorously preserving the privacy, integrity (e.g., blockchain integration, federated trust learning), and confidentiality of the entire system.

CONFLICTS OF INTEREST

We wish to confirm that there are no known conflicts of interest associated with this publication and there has been no significant financial support for this work that could have influenced its outcome.

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APPENDIX

Table 11. Abbreviations

Abbreviated	Name	Abbreviated	Name
ABC	Artificial Bee Colony	**ML-HSOR**	Multi-Level Hierarchical Secure and Optimal Routing
AEC	Average Energy Consumption	**MN**	Malicious Node
AHbCOA	Artificial Hummingbird Cheetah Optimizer Algorithm	**MND-DTM***	Malicious Node Detection using Dynamic Trust Management
Bi-LSTM	Bidirectional Long Short-Term Memory	**NF-SSOA**	Neuro-Fuzzy-based clustering along with Sparrow Search Optimization Algorithm
BS	Base Station	**NFV**	Node Fitness Value
CH	Cluster Head	**OTV**	Overall Trust Values
CHSTF*	Cluster Head Selection using Trust Function	**PL-COA**	Polarity Learning-based Chimp Optimization Algorithm
CI	Contextual Information	**PTS**	path trust scores
CL	Cluster Leader	**QLAMSR**	Query-based Location-Aware Secure Multicast Routing for Wireless Sensor Networks
CLS FTCM	Cross Layer Security Based Fuzzy Trust Calculation Mechanism	**QoS**	Quality of Service
CMBO	Cat-and-Mouse-Based Optimizer	**QPC**	Quality of Peer-to-peer Communication
DL	Deep Learning	**RC6**	Rivest Cipher 6
DoS	Denial of Service	**RPL**	Routing Protocol for Low-power and Lossy networks
DSMND	Dual-Stage Malicious Node Detection	**SCFTO**	Secure Clustering protocol with Fuzzy Trust evaluation and Outlier detection
DST	Dempster–Shafer theory	**SCHs**	Secure Cluster Heads
DST-WOA	Dempster-Shafer Theory with Whale Optimization Algorithm	**SEAT-DSR**	Spatial and Energy Aware Trusted dynamic Distance Source Routing
EATSRA	Energy Aware Trust Based Secure Routing Algorithm	**SEP**	Stable Election Protocol
ECC	Elliptic Curve Cryptography	**SET-SCHNORR**	Setup SCHNORR
ECHSA	efficient cluster head selection algorithm	**SET- ìTESLA**	Setup ìTESLA
EESTCA	Energy Efficient Secure Trust based Clustering Algorithm	**SI-BBMO**	Self-Improving Bumble Bee Mating Optimization
EGSCFO	Evolutionary Game based Secure Clustering protocol with Fuzzy trust evaluation and Outlier detection	**SQEER**	Secured QoS aware Energy Efficient Routing
FL	Federated Learning	**TA**	Trust Aggregation
ETDMA*	Efficient Trust-based Decision-Making Approach	**TADA**	trust-based attack detection algorithm

continued on following page

Table 11. Continued

Abbreviated	Name	Abbreviated	Name
ETERS	Energy aware Trust-based Efficient Routing Scheme	**TAGA**	Trust-based Adaptive Genetic Algorithm
FDTM-IoT	Fuzzy-based Dynamic Trust Management for IoT	**TBSEER**	Trust Based Secure and Energy Efficient Routing
FTM	Fuzzy Trust Model	**TC**	Trust Composition
FTM-ABC	Fuzzy Trust Model and the ABC algorithm	**T-CBRSS**	Trust-based Clustering and Best Route Selection Strategy
IFTCSR*	Intuitionistic Fuzzy-based Trust Computation for Secure Routing	**TE**	Trust Establishment
IoT	Internet of Things	**TEEFCA**	Trusted Energy Efficient Fuzzy logic based Clustering Algorithm
IT2-FLC	Interval Type-2 Fuzzy Logic Controller	TMS	Trust Management Systems
ITU-T	International Telecommunication Union Telecommunication Standardization Sector	**T-MMFO**	Trust-based Modified Moth Flame Optimization
LOA	Lion Optimization Algorithm	**TOPSIS**	Technique for Order of Preference by Similarity to Ideal Solution
LLT	Low Latency Time	**TP**	Trust Propagation
LTS	Large scale Trust Scheme	**TPM**	Trusted Platform Module
LWTM	Light Weight Trust Mechanism	**TSEEFC***	Trust aware Secured Energy Efficient Fuzzy Clustering-based protocol
MAF-DRL	Multi-Agent Framework for Deep Reinforcement Learning	**TTES**	TPM-based Trust Enhancement Scheme
MARL	Multi-Agent Deep Reinforcement Learning	**WSNs**	Wireless Sensor Networks
MFO	Moth Flame Optimization	**WWSNs**	Weak-link Wireless Sensor Networks
ML	Machine Learning		

***** The acronyms indicated with * are recommended by the authors of this article.

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